

KAUSHIK BORA

Borlaug Institute for South Asia (BISA-CIMMYT), NASC Complex, New Delhi – 110012, India

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PROFESSIONAL PROFILE

Researcher in the field of agricultural and development economics. Experienced in climate and policy research related to climatic impacts, adaptation, and policy analysis. Skilled in econometric applications, meta-analysis, and data analysis, with several years of project management and partnership experience with governmental and development-focused institutions. Experience in collaborating with institutions of South Asia in the agriculture sector and extensive work focused on several parts of India.

RESEARCH INTERESTS

Agricultural and Development Economics · Environmental Economics · Climate Change Economics

PROFESSIONAL EXPERIENCE

Agricultural Economist (Research Associate)

June 2022 – Present

Borlaug Institute for South Asia (BISA-CIMMYT), New Delhi

- Involved with the *Atlas of Climatic Adaptation in South Asian Agriculture (ACASA)* and *Next Generation Bundled Crop Insurance in South Asia*.
- Use of high-resolution gridded data for climatic risk assessment.
- Application of spatial econometric models for adaptation effectiveness.
- Leading a team of socio-economists on the suitability and scalability of adaptation options.
- Mixed methods research for stakeholder prioritisation and gender-responsive scaling.
- Imparting training on climate risk management, meta-analysis and responsible scaling in agriculture.
- Development of a climate-smart agriculture index for policy prioritisation.

Visiting Scientist

October – November 2024

Evans School of Policy Analysis and Research, University of Washington, Seattle

- Work on cost-benefits and gender suitability of adaptation options.
- Application of spatial econometric models for adaptation effectiveness.

Project Manager, Climate Change

January – May 2022

MS Swaminathan Research Foundation, Chennai

- Led monitoring of projects related to climate change.
- Impact assessment of climate change in agriculture for major crops in Odisha and Tamil Nadu.
- Development of a composite index indicator for the climate change adaptation project.

Research Fellow

July 2017 – January 2022

Economic Analysis Unit (EAU), Indian Statistical Institute, Bangalore

- Doctoral research on the effect of climatic extremes on fertilizer use in Indian agriculture using panel econometric techniques.
- Capacity building of stakeholders and research collaborators in workshops.

Technical Assistant

September 2016 – April 2017

Office of Vice-Chairperson, Kerala State Planning Board, Thiruvananthapuram

EDUCATION

- 2015–22 **PhD in Development Studies**
Indian Statistical Institute, Bangalore
(Degree awarded by Tata Institute of Social Sciences, Mumbai)
Thesis: A Study of Climatic Factors and Fertilizer Use in Indian Agriculture
- 2013–15 **M.Sc. Agricultural Economics**
Punjab Agricultural University, Ludhiana
Thesis: Productivity Analysis of Rice and Wheat in Indo-Gangetic Plains of India
- 2009–13 **B.Sc. in Agriculture**
Assam Agricultural University, Jorhat

PUBLICATIONS

1. Aggarwal, P., Shirsath, P., Gangopadhyay, P., Rahman, H., Rao, C.A.R., Ojha, R.B., Rathnayake, U.K., Barik, A., Deo, A., **Bora, K.**, Bandyopadhyay, S., Singh, S., Goswami, U.P., Pragya, P., Gupta, R. (2026). A web-based, gridded Atlas of multi-hazard, multi-species climate risks and technology options to support adaptation planning in South Asian agriculture. *Climate Services*, 42, 100671. <https://doi.org/10.1016/j.cliser.2026.100671> (Elsevier)
2. Maiti, S., Panja, A., Deo, A., **Bora, K.**, Garai, S., Shirsath, P., and Aggarwal, P.K. (2026). Climate adaptation in South Asian livestock production systems: Emerging trends and contextual insights. *Tropical Animal Health and Production*, 58(4), 233. <https://doi.org/10.1007/s11250-026-05030-9> (Springer Nature)
3. Gupta, R., **Bora, K.**, Gangopadhyay, P., and Aggarwal, P.K. (2026). Who adopts what? A systematic global review of gender dynamics in climate-smart agriculture. *Agricultural Systems*, 234, 104688. <https://doi.org/10.1016/j.agsy.2026.104688> (Elsevier)
4. **Bora, K.**, Shirsath, P.B., and Aggarwal, P. (2025). Hotspots of climate smart agriculture and climatic risks in India. *Regional Environmental Change*, 121(25). <https://doi.org/10.1007/s10113-025-02456-z> (Springer Nature)
5. Jain, N., Verma, H., Deo, A., **Bora, K.** et al. (2025). Effect of climate change adaptation options on maize yield across different agro-climatic zones in South Asia: A meta-analysis. *Agronomy for Sustainable Development*, 45(78). <https://doi.org/10.1007/s13593-025-01075-6> (Springer Nature)
6. Kumari, V.V., Deo, A., **Bora, K.**, Rao, C.R., Raju, B.M.K., Chandran, M.S. et al. (2025). Adapting sorghum and other millets to climate challenges: An integrated bibliometric and meta-analysis of global literature. *Journal of Agriculture and Food Research*, 21, 101987. [10.1016/j.jafr.2025.101987](https://doi.org/10.1016/j.jafr.2025.101987) (Elsevier)
7. Ansari, M.A., Joshi, H., Mehta, P., Deo, A., **Bora, K.**, Rao, C.R. et al. (2025). Climate-smart and risk-resilient adaptation strategies for potato production: A meta-analysis for South Asia. *Environmental Challenges*, 101183. <https://doi.org/10.1016/j.envc.2025.101183> (Elsevier)
8. **Bora, K.** (2022). Rainfall shocks and fertilizer use: A district level study of India. *Environment and Development Economics*, 1–22. <https://doi.org/10.1017/S1355770X21000413> (Cambridge University Press)
9. **Bora, K.** (2022). Spatial patterns of fertilizer use and imbalances: Evidence from rice cultivation in India. *Environmental Challenges*, 27, 100452. <https://doi.org/10.1016/j.envc.2022.100452> (Elsevier)
10. **Bora, K.**, Bhattacharya, S., and Niyati, S. (2018). Paddy prices and marketing agencies in India: An empirical analysis of effectiveness of public procurement. *Agricultural Economics Research Review*, 31(Conf. Spl.), 157–166. [10.22004/ag.econ.281153](https://doi.org/10.22004/ag.econ.281153)

WORKING PAPERS

1. **Bora, K.**, Gupta, R., Gangopadhyay, P.K., Rama Rao, C.A., Raju, B.M.K., Sapkota, B., and Aggarwal, P.K. Gender responsive technology assessment index: Scaling inclusive climate smart interventions in data-limited settings. (Under Review). [Preprint Link]
2. Maiti, S., **Bora, K.**, Zade, S., Yadav, B., Sanjyal, S., Shirsath, P., Garai, S., Runa, R.A., Weerasinghe, W.M.P.B., Thornton, P., and Aggarwal, P.K. Identifying climatic risk hotspots for the dairy production system of South Asia: An integrated analysis with projections for 2050s and 2080s. (Under Review). [Preprint Link]
3. **Bora, K.** and Rahut, D. How farmers adjust irrigation during climate extremes? Insights from rice farming in India. (Under Review).
4. **Bora, K.**, Mkondiwa, M., and Aggarwal, P.K. Targeted strategies for closing yield gaps to achieve food self-sufficiency under current and future demand scenarios: Evidence from India. (Under Review).
5. **Bora, K.** and Aggarwal, P.K. Integrating yield gaps and sustainable intensification for a spatially explicit prioritization framework in agriculture. (Under Review).
6. **Bora, K.** Dynamic adjustments in fertilizer use under weather shocks: Micro-level evidence from India. (Under Review).
7. Gangopadhyay, P.K., **Bora, K.**, Bandyopadhyay, S.K. and Aggarwal, P.K. Constraint interactions govern the scalability of climate adaptation. *Under Review*.
8. **Bora, K.** and Niyati, S. Trends in female labour absorption in rice cultivation: Evidence from India. Paper presented at the 31st Virtual Conference of the International Conference of Agricultural Economists. [Preprint Link]

SEMINAR AND CONFERENCE PRESENTATIONS

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| 2025 | Gender Responsive Technology Assessment Index: Scaling Inclusive Climate Smart Interventions in Data-Limited Settings. ANAPRI–World Bank Stakeholder Conference on “Leveraging the Agri Food System for Jobs”, Kigali, Rwanda, November 3-5. |
| 2024 | Yield Gap Decomposition and Technical Inefficiencies Under Climatic Risks in Monsoon Rice (poster). 32nd International Conference of Agricultural Economists (ICAE), August 2–7. |
| 2021 | Spatial Patterns of Fertilizer Use and Imbalances: Evidence from Rice Cultivation in India. ADBI Conference on Environmental Challenges and Agricultural Sustainability in Asia, December 8–10. |
| 2021 | Effects of Rainfall Shocks on Chemical Fertilizer Application: A Plot Level Study of Paddy in India. 15th Annual Meeting, Environment for Development (EfD) Initiative, November 15–19. |
| 2021 | Rainfall Shocks and Fertilizer Use: A District Level Study of India (poster). 6th International Climate Change Adaptation Conference, Adaptation Futures 2020, October 5–8. |
| 2019 | Fertilizer Use and Rainfall Shocks in Indian Agriculture. Tenth Biennial Conference, Indian Society of Ecological Economics, CESS, Hyderabad, November 6–8. |

SOFTWARE KNOWLEDGE

Stata (Proficient) · R (Proficient) · Python (Beginner) · QGIS (Beginner)

TEACHING EXPERIENCE

- 2016 Teaching Assistant, Indian Statistical Institute, Bangalore
Course: Research Methodology for Development Studies
- 2021 Teaching Assistant, Indian Statistical Institute, Bangalore
Course: Poverty and Inequality
- 2022 Guest Faculty, Indian Statistical Institute, Bangalore
Course: Econometrics for Social Science
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Last updated: May 23, 2026